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# ELECTRONIC DEVICE, METHOD, AND NON-TRANSITORY COMPUTER-READABLE STORAGE MEDIUM FOR HANDOVER OF AUDIO SERVICE

## Abstract

An electronic device is provided. The electronic device is configured to, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via a speaker, while the first audio service is provided, receive, from the first external electronic device, a request that ceases the synchronizing to the BIS and maintains the synchronizing to the periodic advertisement, and, while providing a second audio service based on data received from the first external electronic device, cease the synchronizing to the BIS and maintain the synchronizing to the periodic advertisement.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365(c), of an International application No. PCT/KR2024/008520, filed on Jun. 20, 2024, which is based on and claims the benefit of a Korean patent application number 10-2023-0107696, filed on Aug. 17, 2023, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0116551, filed on Sep. 1, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND
1. Field
[0002] The disclosure relates to an electronic device, a method, and a non-transitory computer-readable storage medium for handover of an audio service.
2. Description of Related Art
[0003] Compared to legacy Bluetooth® (or classic Bluetooth), Bluetooth® low energy (BLE) may provide reduced power consumption and provide at least a similar or often greater communication range between connected devices. BLE may be provided on an industrial, scientific, and medical (ISM) radio band.
[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.
SUMMARY
[0005] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device, a method, and a non-transitory computer-readable storage medium for handover of an audio service.
[0006] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
[0007] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes communication circuitry for Bluetooth low energy (BLE) communication, a speaker, memory, including one or more storage media, storing instructions, and at least one processor communicatively coupled to the communication circuitry, the speaker, and the memory, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via the speaker, while the first audio service is provided, receive, via a communication link between the electronic

device and the first external electronic device, from the first external electronic device, a request that ceases the synchronizing to the BIS and maintains the synchronizing to the periodic advertisement, using the communication circuitry, and, while providing a second audio service based on data received via the communication link from the first external electronic device, cease the synchronizing to the BIS and maintain the synchronizing to the periodic advertisement.

[0008] In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes communication circuitry for Bluetooth low energy (BLE) communication, a speaker, memory, including one or more storage media, storing instructions, and at least one processor communicatively coupled to the communication circuitry, the speaker, and the memory, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via the speaker, while the first audio service is provided, receive, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a signal for starting a second audio service, using the communication circuitry, in response to the signal, cease providing the first audio service based on data for the BIS broadcasted from the second external electronic device, and, based on the signal, maintain the synchronizing to the periodic advertisement, while ceasing providing the first audio service and providing the second audio service based on data received via the communication link from the first external electronic device.

[0009] In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes communication circuitry for Bluetooth low energy (BLE) communication, memory, including one or more storage media, storing instructions, and at least one processor communicatively coupled to the communication circuitry and the memory, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to transmit, to another electronic device via a communication link between the electronic device and the other electronic device, first information on a periodic advertisement of an external electronic device, using the communication circuitry, receive, via the communication link, from the other electronic device, second information indicating a state of the other electronic device synchronized to a broadcast isochronous stream (BIS) from the external electronic device in accordance with the first information, using the communication circuitry, after the second information is received, detect an event for starting an audio service provided via the communication link, in conjunction with the other electronic device, and, in response to the event, transmit, to the other electronic device via the communication link, a request that ceases the synchronizing to the BIS and maintains the synchronizing of the periodic advertisement, using the communication circuitry.

[0010] In accordance with another aspect of the disclosure, a method is provided. The method is executed in an electronic device with communication circuitry for Bluetooth low energy (BLE) communications and a speaker. The method includes, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, providing a first audio service outputting audio via the speaker, while the first audio service is provided, receiving, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a request that ceases the synchronizing to the BIS and maintains the synchronizing to the periodic advertisement, using the communication circuitry, and, while providing a second audio service based on data received via the communication link from the first external electronic device, ceasing the synchronizing to the BIS and maintaining the synchronizing to the periodic advertisement.

[0011] In accordance with another aspect of the disclosure, a method is provided. The method is executed in an electronic device with communication circuitry for Bluetooth low energy (BLE) communication and a speaker. The method includes, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, providing a first audio service outputting audio via the speaker, while the first audio service is provided, receiving, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a signal for starting a second audio service, using the communication circuitry, in response to the signal, ceasing providing the first audio service based on data for the BIS broadcasted from the second external electronic device, and, based on the signal, maintaining the synchronizing to the periodic advertisement, while ceasing providing the first audio service and providing the second audio service based on data received via the communication link from the first external electronic device.

[0012] In accordance with another aspect of the disclosure, a method is provided. The method is executed in an electronic device with communication circuitry for Bluetooth low energy (BLE) communication. The method includes transmitting, to another electronic device via a communication link between the electronic device and the other electronic device, first information on a periodic advertisement of an external electronic device, using the communication circuitry, receiving, via the communication link, from the other electronic device, second information indicating a state of the other electronic device synchronized to a broadcast isochronous stream (BIS) from the external electronic device in accordance with the first information, using the communication circuitry, after the second information is received, detecting, an event for starting an audio service provided via the communication link, in conjunction with the other electronic device, and, in response to the event, transmitting, to the other electronic device via the communication link, a request that ceases the synchronizing to the BIS and maintains the synchronizing of the periodic advertisement, using the communication circuitry.

[0013] In accordance with another aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by at least one processor of an electronic device individually or collectively, cause the electronic device to perform operations are provided. The operations include, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, providing a first audio service outputting audio via a speaker, while the first audio service is provided, receiving, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a request that ceases the synchronizing to the BIS and maintains the synchronizing to the periodic advertisement, using a communication circuitry, and, while providing a second audio service based on data received via the communication link from the first external electronic device, ceasing the synchronizing to the BIS and maintaining the synchronizing to the periodic advertisement.

[0014] In accordance with another aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by at least one processor of an electronic device individually or collectively, cause the electronic device to perform operations are provided. The operations include, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via the speaker, while the first audio service is provided, receive, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a signal for starting a second audio service, using the communication circuitry, in response to the signal, cease providing the first audio service based on data for the BIS broadcasted from the second external electronic device, and, based on the signal, maintain the synchronizing to the periodic advertisement, while ceasing providing the first audio service and providing the second audio service based on data received via the communication link from the first external electronic device.

[0015] In accordance with another aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by at least one processor of an electronic device individually or collectively, cause the electronic device to perform operations are provided. The operations include, transmit, to another electronic device via a communication link between the electronic device and the other electronic device, first information on a periodic advertisement of an external electronic device, using the communication circuitry, receive, via the communication link, from the other electronic device, second information indicating a state of the other electronic device synchronized to a broadcast isochronous stream (BIS) from the external electronic device in accordance with the first information, using the communication circuitry, after the second information is received, detect, an event for starting an audio service provided via the communication link, in conjunction with the other electronic device, and, in response to the event, transmit, to the other electronic device via the communication link, a request that ceases the synchronizing to the BIS and maintains the synchronizing of the periodic advertisement, using the communication circuitry.

[0016] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0018] FIG. **1** illustrates an example of an environment including an electronic device, a first external electronic device, and a second external electronic device according to an embodiment of the disclosure;

[0019] FIG. **2** is a simplified block diagram of a first external electronic device according to an embodiment of the disclosure;

[0020] FIG. **3** is a simplified block diagram of an electronic device according to an embodiment of the disclosure;

[0021] FIG. **4** illustrates a method of maintaining a periodic advertisement based on a request caused in accordance with a first handover of an audio service according to an embodiment of the disclosure;

[0022] FIG. **5** illustrates a method of executing in an electronic device in accordance with synchronizing to a broadcast isochronous stream (BIS) according to an embodiment of the disclosure;

[0023] FIG. **6** illustrates a method executed in a first external electronic device in accordance with detection of an event according to an embodiment of the disclosure;

[0024] FIGS. **7** and **8** illustrate a method executed in an electronic device in accordance with reception of a request according to various embodiments of the disclosure;

[0025] FIG. **9** illustrates a method of resuming synchronizing to a BIS based on another request caused for a second handover of an audio service according to an embodiment of the disclosure;

[0026] FIG. **10** illustrates a method executed in a first external electronic device in accordance with

detection of another event according to an embodiment of the disclosure;
[0027] FIG. **11** illustrates a method of maintaining a periodic advertisement based on a signal caused for a first handover of an audio service according to an embodiment of the disclosure;
[0028] FIG. **12** illustrates a method of resuming synchronizing to a BIS based on another signal caused for a second handover of an audio service according to an embodiment of the disclosure; and
[0029] FIG. **13** is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure.
[0030] Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.
DETAILED DESCRIPTION
[0031] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.
[0032] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.
[0033] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.
[0034] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.
[0035] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless fidelity (Wi-Fi) chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a finger-print sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.
[0036] FIG. **1** illustrates an example of an environment including an electronic device, a first external electronic device, and a second external electronic device according to an embodiment of the disclosure.
[0037] Referring to FIG. **1**, an environment **100** may include an electronic device **101**, a first external electronic device **111**, and a second external electronic device **112**.
[0038] The electronic device **101** in the environment **100** may provide an audio service in conjunction with the first external electronic device **111**. As a non-limiting example, the electronic device **101** may be a sink device, and the first external electronic device **111** may be a source

device. As a non-limiting example, the electronic device **101** may be a source device.
[0039] For example, the electronic device **101** may be configured to output audio. For example, the electronic device **101** may output audio based on data received from the first external electronic device **111** via a communication link **131** between the electronic device **101** and the first external electronic device **111**. For example, the communication link **131** may include an asynchronous connectionless link (ACL), which is a control data link. As a non-limiting example, the communication link **131** may be used for a connected isochronous stream (CIS).
[0040] For example, the electronic device **101** may be configured to obtain audio. For example, the electronic device **101** may obtain audio, based on a request received from the first external electronic device **111** via the communication link **131**. For example, the electronic device **101** may transmit, to the first external electronic device **111** via the communication link **131**, information on the obtained audio, based on data received from the first external electronic device **111** via the communication link **131**.
[0041] The electronic device **101** may provide an audio service based on data broadcasted from the second external electronic device **112**. As a non-limiting example, the electronic device **101** may be a sink device and the second external electronic device **112** may be a source device.
[0042] For example, the electronic device **101** may output audio, based on data for a broadcast isochronous stream (BIS) from the second external electronic device **112**. For example, the audio may be broadcasted via a broadcast isochronous group (BIG). For example, the BIG may include multiple instances for the BIS. For example, an event of the BIG may include one or more BIS events. For example, each of the one or more BIS events may include one or more sub-events. For example, the audio may be outputted based on synchronizing to the BIS from the second external electronic device **112**, as indicated by arrow **161**. For example, the synchronizing to the BIS may indicate receiving the data for the BIS. For example, the synchronizing to the BIS may be executed based on synchronizing to a periodic advertisement **142** of the second external electronic device **112**.
[0043] The first external electronic device **111** within the environment **100** may be used to assist the electronic device **101** in synchronizing to the BIS. For example, the first external electronic device **111** may include at least a portion of the electronic device **1301** of FIG. **13**, or may correspond to at least a portion of the electronic device **1301** of FIG. **13**.
[0044] As a non-limiting example, unlike the electronic device **101**, the first external electronic device **111** may include a display. For example, the first external electronic device **111** may display, on the display, a user interface for checking, identifying, or determining whether to output audio from the electronic device **101** based on data broadcasted from the second external electronic device **112** (e.g., the data for the BIS). For example, the first external electronic device **111** may be used to receive a user input for the user interface. For example, the user input may be used to indicate outputting the audio. For example, the first external electronic device **111** may assist the electronic device **101** in synchronizing to the BIS through operations related to receiving the user input.
[0045] As a non-limiting example, the first external electronic device **111** may include a battery having a capacity greater than a capacity of a rechargeable battery within the electronic device **101**. For example, in order to reduce power consumption of the rechargeable battery within the electronic device **101**, the first external electronic device **111** may perform a scanning for one or more advertisements **141** executed by the second external electronic device **112**, instead of the electronic device **101**. For example, the one or more of the advertisements **141** may indicate a periodic advertisement **142** executed by the second external electronic device **112**, as indicated by arrows **151**. For example, at least one packet advertised from the second external electronic device **112** according to the one or more advertisements **141** may include information on the periodic advertisement **142**. For example, the periodic advertisement **142** may indicate resources for broadcasting (**143**) the data for the BIS, as indicated by arrows **152**. For example, at least one

packet advertised from the second external electronic device **112** according to the periodic advertisement **142** may include information on broadcasting (**143**) the data for the BIS (e.g., information on resources for broadcasting (**143**) the data for the BIS). For example, the first external electronic device **111** may assist, through operations related to the scanning, the electronic device **101** in synchronizing to the BIS, as indicated by arrow **161**. For example, the first external electronic device **111** may transmit, to the electronic device **101**, information indicating a result of the scanning (e.g., information on the periodic advertisement **142**). For example, the information may be transmitted to the electronic device **101** via the communication link **131**. However, it is not limited thereto. For example, the information may be transmitted from the first external electronic device **111** to the electronic device **101** via a device (e.g., cradle) for storing the electronic device **101** and charging the rechargeable battery of the electronic device **101**.

[0046] The second external electronic device **112** within the environment **100** may broadcast the data for the BIS. Although not illustrated in FIG. **1**, the second external electronic device **112** may execute one or more advertisements **141** to provide information for accessing the data for the BIS. For example, the one or more advertisements **141** may provide information for accessing to the periodic advertisement **142**. For example, the second external electronic device **112** may execute the periodic advertisement **142** to provide information for accessing the data for the BIS. For example, the periodic advertisement **142** may provide information for accessing the data for the BIS broadcasted from the second external electronic device **112**. For example, the electronic device **101** may synchronize to the BIS by accessing the data for the BIS (or receiving the data for the BIS).

[0047] As a non-limiting example, the environment **100** may further include another electronic device **102**. For example, the other electronic device **102** may be a device paired with the electronic device **101**. As a non-limiting example, the electronic device **101** may be worn on a user's right ear and the other electronic device **102** may be worn on a user's left ear. As a non-limiting example, the electronic device **101** paired with the other electronic device **102** may indicate that the electronic device **101** and the other electronic device **102** are configured as one set to provide an audio service. As a non-limiting example, the electronic device **101** paired with the other electronic device **102** may indicate that the electronic device **101** and the other electronic device **102** are connected via a communication link **132**. As a non-limiting example, the electronic device **101** paired with the other electronic device **102** may indicate that the electronic device **101** and the other electronic device **102** are used for stereophonic sound. For example, audio outputted via a speaker of the electronic device **101** may correspond to audio outputted via a speaker of the other electronic device **102**. For example, audio obtained via a microphone of the electronic device **101** may correspond to audio obtained via a microphone of the other electronic device **102**.

[0048] For example, the other electronic device **102** may provide an audio service in conjunction with the first external electronic device **111** and the electronic device **101**. As a non-limiting example, the other electronic device **102** may be a sink device. As a non-limiting example, the other electronic device **102** may be a source device.

[0049] For example, the other electronic device **102** may be configured to output audio. For example, the other electronic device **102** may output audio based on data received from the first external electronic device **111** via a communication link **133** between the other electronic device **102** and the first external electronic device **111**. For example, the communication link **133** may include an ACL, which is a control data link. As a non-limiting example, the communication link **133** may be used for CIS. For example, the other electronic device **102** may output audio based on data received from the first external electronic device **111** via the electronic device **101** (e.g., data received via the communication link **131** and the communication link **132**).

[0050] For example, the other electronic device **102** may be configured to obtain audio. For example, the other electronic device **102** may obtain audio based on a request received from the first external electronic device **111** via the communication link **133**. For example, the other electronic device **102** may obtain audio based on a request received from the first external electronic device **111** via the electronic device **101** (e.g., a request received via the communication link **131** and the communication link **132**). For example, the other electronic device **102** may transmit, to the first external electronic device **111** via the communication link **133**, information on the obtained audio, based on data received from the first external electronic device **111** via the communication link **133**. For example, the other electronic device **102** may transmit, to the first external electronic device **111** via the electronic device **101**, information on the obtained audio, based on data received from the first external electronic device **111** via the electronic device **101** (e.g., data received via the communication link **131** and the communication link **132**). For example, the information may be transmitted to the first external electronic device **111**, via the communication link **131** and the communication link **132**.

[0051] For example, the other electronic device **102** may provide an audio service, based on data broadcasted from the second external electronic device **112**. As a non-limiting example, the other electronic device **102** may be a sink device.

[0052] For example, the other electronic device **102** may output audio based on the data for the BIS from the second external electronic device **112**. For example, the other electronic device **102** may output, while the electronic device **101** outputs audio via one or more first BIS events within an event of the BIG, audio via one or more second BIS events within the event of the BIG. For example, the audio outputted from the other electronic device **102** may be outputted based on synchronizing to the BIS from the second external electronic device **112**, as indicated by arrow **162**. For example, synchronizing to the BIS may indicate receiving the data for the BIS. For example, synchronizing to the BIS may be executed based on synchronizing to a periodic advertisement **142** of the second external electronic device **112**.

[0053] As a non-limiting example, the other electronic device **102** may be configured to execute at least a portion of operations of the electronic device **101** to be exemplified below.

[0054] As a non-limiting example, the electronic device **101** may provide a first audio service based on signaling from the second external electronic device **112**, and may provide a second audio service based on signaling with the first external electronic device **111**. As a non-limiting example, providing simultaneously the first audio service and the second audio service may be limited or disabled within the electronic device **101**. For example, in a case that both the first audio service and the second audio service include outputting audio, outputting audio for the first audio service within the electronic device **101** may not be executed simultaneously with outputting audio for the second audio service within the electronic device **101**.

[0055] For example, a priority of the first audio service may be lower than a priority of the second audio service. As a non-limiting example, the priority of the first audio service and the priority of the second audio service may be set by the first external electronic device **111**. As a non-limiting example, the priority of the first audio service and the priority of the second audio service may also be set by the electronic device **101**.

[0056] As a non-limiting example, the electronic device **101** may detect (or identify) (or recognize) (or check) (or determine) to start providing a second audio service having a higher priority than the priority of the first audio service, while providing the first audio service. For example, the electronic device **101** may execute, based on the detection, a handover (or switching) (or changing) of an audio service provided by using the electronic device **101**. For example, the electronic device **101** may execute a first handover from the first audio service to the second audio service, based on the detection. For example, the first audio service may be ceased in accordance with starting of the second audio service (or completing of the first handover).

[0057] As a non-limiting example, the electronic device **101** may detect to cease (or terminate) providing the second audio service. For example, the electronic device **101** may execute, based on the detection, a handover of an audio service provided using the electronic device **101** to resume the first audio service. For example, the electronic device **101** may execute, based on the detection, a second handover from the second audio service to the first audio service. For example, the first audio service may be resumed in accordance with terminating of the second audio service (or in accordance with completing of the second handover). For example, the second audio service may be terminated at a first timing, and the first audio service may be resumed from a second timing after the first timing.

[0058] For example, as a length of a time interval from the first timing to the second timing increases, the discomfort felt by a user of the electronic device **101** may increase. For example, the length of the time interval may be a parameter indicating performance of the electronic device **101** recognized by the user. For example, reducing the length of the time interval may enhance a quality of an audio service provided using the electronic device **101**.

[0059] In order to reduce the length of the time interval, an electronic device **101** to be exemplified through descriptions below may maintain synchronizing to the periodic advertisement **142** of the second external electronic device **112** while the second audio service is provided. For example, since operations related to executing new synchronizing to the periodic advertisement **142** to resume the first audio service are skipped by maintaining the synchronizing to the periodic advertisement **142**, the electronic device **101** may reduce the length of the time interval. Meanwhile, a first external electronic device **111** to be exemplified through descriptions below may execute an operation to assist in reducing the length of the time interval.

[0060] For example, the electronic device **101** may include components used to reduce the length of the time interval. For example, the first external electronic device **111** may include components used to reduce the length of the time interval. The components of the first external electronic device **111** are exemplified in FIG. **2**, and the components of the electronic device **101** are exemplified in FIG. **3**.

[0061] FIG. **2** is a simplified block diagram of a first external electronic device according to an embodiment of the disclosure.

[0062] Referring to FIG. **2**, the first external electronic device **111** may include at least one processor **201** and communication circuitry **202**.

[0063] The at least one processor **201** may include at least a portion of the processor **1320** of FIG. **13** or may correspond to at least a portion of the processor **1320** of FIG. **13**. The at least one processor **201** may be configured to control the communication circuitry **202**. The at least one processor **201** may be configured to execute instructions stored in memory (not illustrated in FIG. **2**) to cause the first external electronic device **111** to perform at least a portion of the operations exemplified in the description of FIG. **1**. The at least one processor **201** may be configured to execute instructions stored in the memory to cause the first external electronic device **111** to perform at least a portion of operations exemplified in descriptions of FIG. **3** to FIG. **12**. For example, the memory may include a non-volatile memory. For example, the memory may include at least a portion (e.g., the non-volatile memory **1334**) of the memory **1330** of FIG. **13** or may correspond to the at least a portion of the memory **1330** of FIG. **13**.

[0064] The communication circuitry **202** may support legacy Bluetooth and/or Bluetooth low energy (BLE). For example, the communication circuitry **202** may be used for communication with the electronic device **101**. For example, the communication circuitry **202** may be used for communication with the second external electronic device **112**. For example, the communication circuitry **202** may be used for communication with the other electronic device **102**. For example, the communication circuitry **202** may be used for the first audio service. For example, the communication circuitry **202** may be used for the second audio service.

[0065] The first external electronic device **111** exemplified in the description of FIG. **2** may execute at least a portion of operations exemplified in descriptions of FIGS. **4**, **6**, **9**, **10**, **11**, and **12**. The operations of the first external electronic device **111** to be exemplified in descriptions of FIGS. **4**, **6**, **9**, **10**, **11**, and **12** may be executed, caused, or controlled by the at least one processor **201**.

[0066] FIG. **3** is a simplified block diagram of an electronic device according to an embodiment of

the disclosure.
[0067] Referring to FIG. **3**, an electronic device **101** may include at least a portion of the electronic device **1302** of FIG. **13**, or may correspond to at least a portion of the electronic device **1302** of FIG. **13**. The electronic device **101** may include at least one processor **301**, communication circuitry **302**, and a speaker **303**. The electronic device **101** may further include a microphone **304**.
[0068] The at least one processor **301** may be configured to control the communication circuitry **302**, the speaker **303**, and the microphone **304**. The at least one processor **301** may be configured to execute instructions stored in memory (not illustrated in FIG. **3**) to cause the electronic device **101** to perform at least a portion of the operations exemplified in the description of FIG. **1**. The at least one processor **301** may be configured to execute instructions stored in the memory individually or collectively to cause the electronic device **101** to perform at least a portion of operations exemplified in descriptions of FIGS. **3** to **12**. For example, the instructions may cause the electronic device **101**, when executed by the at least one processor individually or collectively, to perform at least a portion of the operations exemplified in the descriptions of FIGS. **3** to **12**. For example, the memory may include a nonvolatile memory.
[0069] The communication circuitry **302** may support legacy Bluetooth and/or Bluetooth low energy (BLE). For example, the communication circuitry **302** may be used for communication with the first external electronic device **111**. For example, the communication circuitry **302** may be used for communication with the second external electronic device **112**. For example, the communication circuitry **302** may be used for communication with the other electronic device **102**. For example, the communication circuitry **302** may be used for the first audio service. For example, the communication circuitry **302** may be used for the second audio service.
[0070] The speaker **303** may be configured to output audio. For example, the speaker **303** may output audio provided from the first external electronic device **111**. For example, the speaker **303** may output audio provided from the second external electronic device **112**. For example, the speaker **303** may be used for the first audio service. For example, the speaker **303** may be used for the second audio service.
[0071] The microphone **304** may be configured to obtain audio caused around the electronic device **101**. For example, the microphone **304** may be used to obtain audio to be provided to the first external electronic device **111**. For example, the microphone **304** may be used to obtain audio to be provided to the other electronic device **102**.
[0072] The electronic device **101** exemplified in the description of FIG. **3** may execute at least a portion of the operations exemplified in the descriptions of FIGS. **4**, **5**, **7**, **8**, **9**, **11**, and **12**. The operations of the electronic device **101** to be exemplified in the descriptions of FIGS. **4**, **5**, **7**, **8**, **9**, **11**, and **12** may be executed, caused, or controlled by the at least one processor **301**.
[0073] FIG. **4** illustrates a method of maintaining a periodic advertisement based on a request caused in accordance with a first handover of an audio service according to an embodiment of the disclosure.
[0074] Referring to FIG. **4**, in operation **401**, the electronic device **101** and the first external electronic device **111** may establish a communication link **131**.
[0075] As a non-limiting example, the communication link **131** may be established, based on an input to a user interface displayed on a display of the first external electronic device **111** in accordance with an advertisement of the electronic device **101**. For example, the advertisement of the electronic device **101** may be executed based on an input (e.g., user input) caused to the cradle exemplified in the description of FIG. **1**. For example, the advertisement of the electronic device **101** may be executed based on detecting that the electronic device **101** is gripped by a user.
[0076] As a non-limiting example, the communication link **131** may be established based on an input to a user interface displayed on the display of the first external electronic device **111** in accordance with a signal (or packet) transmitted (or broadcasted) (or advertised) from the electronic device **101** in accordance with an advertisement of the first external electronic device

**111**.

[0077] In operation **402**, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, information on the periodic advertisement **142** of the second external electronic device **112**.

[0078] For example, the second external electronic device **112** may execute one or more advertisements **141** (not illustrated in FIG. **4**). For example, the one or more advertisements **141** may indicate advertising at least one first packet (e.g., an ADV_EXT_IND packet) on each of primary advertising channels (e.g., Ch. 37, Ch. 38, and Ch. 39). For example, the at least one first packet may indicate providing additional data via another (or additional) advertisement (e.g., extended advertisement). For example, the at least one first packet may indicate at least one second packet (e.g., an AUX_ADV_IND packet) (or extended advertisement packet). For example, the first external electronic device **111** may receive, based on the at least one first packet received from the second external electronic device **112**, from the second external electronic device **112**, at least one second packet, which is advertised on a portion of secondary advertising channels different from the primary advertising channels and indicates the periodic advertisement **142** (e.g., operation **403**). For example, the first external electronic device **111** may obtain, based on the at least one second packet, the information (e.g., logical link (LL)_PERIODIC_SYNC_IND) on the periodic advertisement **142**, and transmit, to the electronic device **101** via the communication link **131**, the information on the periodic advertisement **142**.

[0079] As a non-limiting example, the first external electronic device **111** may, in response to an input to a user interface displayed on the display of the first external electronic device **111** in accordance with reception of the at least one first packet, obtain the information on the periodic advertisement **142** based on the at least one second packet. For example, the input may be caused to output audio via the electronic device **101**, based on data for a BIS from the second external electronic device **112** (e.g., operation **404**).

[0080] For example, the information on the periodic advertisement **142** may indicate an interval of the periodic advertisement **142** of the second external electronic device **112**, a hopping sequency of the periodic advertisement **142** of the second external electronic device **112**, and/or an access address for synchronizing to the BIS from the second external electronic device **112**.

[0081] The electronic device **101** may receive, from the first external electronic device **111** via the communication link **131**, the information on the periodic advertisement **142**.

[0082] In operation **403**, the second external electronic device **112** may execute the periodic advertisement **142**. For example, the second external electronic device **112** may execute the periodic advertisement **142** by advertising at least one packet. For example, the at least one packet may include information on an interval of the periodic advertisement **142** of the second external electronic device **112**, information on an offset of a transmission in accordance with the periodic advertisement **142** of the second external electronic device **112**, information on a channel map of the periodic advertisement **142** of the second external electronic device **112**, information on the ACL, information on an initial value of cyclic redundancy check (CRC), information on a counter (e.g., PeriodicEventCounter) of an event of the periodic advertisement **142** of the second external electronic device **112**, and information (or advertisement data) on a broadcast isochronous group (BIG) (e.g., including multiple instances for the BIS from the second external electronic device **112**) scheduled (or generated) (or obtained) by the second external electronic device **112**. For example, the periodic advertisement **142** may be used for synchronizing to the BIS from the second external electronic device **112**, as indicated by arrow **152**.

[0083] In operation **404**, the second external electronic device **112** may execute broadcasting (**143**) data for the BIS indicated by the periodic advertisement **142**. For example, indicating that the periodic advertisement **142** broadcasts (**143**) the data for the BIS may mean (or indicate) that at least one packet advertised in accordance with the periodic advertisement **142** includes information on broadcasting (**143**) the data for the BIS. For example, the data for the BIS may be used to output

audio.
[0084] In operation **405**, the electronic device **101** may synchronize to the periodic advertisement **142**, based on the information on the periodic advertisement **142** received in operation **402**. For example, the synchronizing to the periodic advertisement **142** may include receiving at least one packet (e.g., the at least one packet exemplified in a description of operation **403**) advertised from the second external electronic device **112** in accordance with the periodic advertisement **142**. For example, the synchronizing to the periodic advertisement **142** may include recognizing or identifying resources for broadcasting data for the BIS from the second external electronic device **112** (e.g., the data for the BIS exemplified in a description of operation **404**).
[0085] In operation **406**, the electronic device **101** may synchronize to the BIS from the second external electronic device **112**, as indicated by arrow **161**, in accordance with the synchronizing to the periodic advertisement **142** executed in operation **405**. For example, the synchronizing to the BIS may include receiving data for the BIS broadcasted from the second external electronic device **112** (e.g., the data for the BIS as exemplified in the description of operation **404**).
[0086] As a non-limiting example, the electronic device **101** may inform (or expose), to the first external electronic device **111**, completion (or success) of the synchronizing to the BIS. For example, since the first external electronic device **111**, which is a device including a display, may be used to control the electronic device **101**, the electronic device **101** may inform the first external electronic device **111** of completion (or success) of the synchronizing of the electronic device **101** to the BIS. For example, since the first external electronic device **111** may be used to control the electronic device **101**, the electronic device **101** may inform that the first audio service to be exemplified in operation **407** is provided and/or that the first audio service to be exemplified in operation **407** is provided. This operation of the electronic device **101** is exemplified in a description of FIG. **5**.
[0087] FIG. **5** illustrates a method of executing in an electronic device in accordance with synchronizing to a broadcast isochronous stream (BIS) according to an embodiment of the disclosure.
[0088] Referring to FIG. **5**, in operation **501**, the electronic device **101** may synchronize to the BIS from the second external electronic device **112**, based on the synchronizing to the periodic advertisement **142** (e.g., operation **405**). Operation **501** may correspond to operation **406** of FIG. **4**.
[0089] In operation **503**, the electronic device **101** may, in response to the synchronizing to the BIS, transmit, to the first external electronic device **111** via the communication link **131**, information indicating that the electronic device **101** is synchronized to the BIS. For example, the information indicating completion (or success) of the synchronizing to the BIS may be transmitted or provided to the first external electronic device **111** via a broadcast receive state of a broadcast audio scan service (BASS). For example, the information indicating completion (or success) of the synchronizing to the BIS may be included in a BIS_Sync field. For example, the information indicating completion (or success) of the synchronizing to the BIS may be included in a metadata field. However, it is not limited thereto. For example, the first external electronic device **111** may, based on reception of the information indicating completion (or success) of the synchronizing to the BIS, recognize that a first audio service to be exemplified in operation **407** is provided by the electronic device **101** and/or that the first audio service to be exemplified in operation **407** will be provided by the electronic device **101**.
[0090] Referring back to FIG. **4**, in operation **407**, the electronic device **101** may provide a first audio service (e.g., the first audio service exemplified in the description of FIG. **1**) that outputs audio via the speaker **303**, based on the synchronizing to the BIS in operation **406**. For example, since the first audio service is provided by a broadcast from the second external electronic device **112** without using a connection between the electronic device **101** and the second external electronic device **112**, the first audio service may also be referred to as a broadcast service.
[0091] In operation **408**, the first external electronic device **111** may transmit, to the electronic

device **101** via the communication link **131**, a request to maintain the synchronizing to the periodic advertisement **142**, while the first audio service is being provided. For example, the request may be configured to cause the electronic device **101** to cease the synchronizing to the BIS from the second external electronic device **112** and maintain the synchronizing to the periodic advertisement **142**. For example, the request may be configured to cause the electronic device **101** to maintain the synchronizing to the periodic advertisement **142**, maintain the synchronizing to the BIS from the second external electronic device **112**, and cease rendering of the data for the BIS from the second external electronic device **112**.

[0092] For example, since the first external electronic device **111** is in a state of recognizing that the first audio service is being provided by the electronic device **101**, based on reception of the information indicating completion (or success) of the synchronizing to the BIS exemplified in the description of FIG. **5**, the first external electronic device **111** may transmit the request. For example, the transmission of the request may be executed based on detecting an event for a start of a second audio service (e.g., the second audio service exemplified in the description of FIG. **1**). The transmission of the request, which is executed in response to the event, may be exemplified in a description of FIG. **6**.

[0093] FIG. **6** illustrates a method executed in a first external electronic device in accordance with detection of an event according to an embodiment of the disclosure.

[0094] Referring to FIG. **6**, in operation **601**, the first external electronic device **111** may receive, from the electronic device **101** via the communication link **131**, the information indicating success of the synchronizing to the BIS. For example, operation **601** may correspond to the counter operation in operation **503** of FIG. **5**. For example, the first external electronic device **111** may recognize, based on the information, that the first audio service is provided in accordance with the BIS from the second external electronic device **112** and/or that the first audio service will be provided in accordance with the BIS from the second external electronic device **112**.

[0095] In operation **602**, the first external electronic device **111** may detect an event for starting the second audio service.

[0096] The event may be caused for the first external electronic device **111**. As a non-limiting example, the event may include an incoming call, an outgoing call, media playback, voice recording, or video recording. As a non-limiting example, the event may be caused based on a signal (e.g., a signal for an incoming call) transmitted from another external electronic device to the first external electronic device **111**. As a non-limiting example, the event may be caused based on a user input (e.g., a user input for starting a voice recording or a user input for starting a video recording) received for the first external electronic device **111**.

[0097] In operation **603**, the first external electronic device **111** may, in response to the event, transmit the request exemplified in the description of operation **408**.

[0098] As a non-limiting example, the first external electronic device **111** may, in accordance with detecting the event, recognize that a priority of the second audio service is higher than a priority of the first audio service, or recognize that a priority of the first external electronic device **111** is higher than a priority of the second external electronic device **112**. For example, the first external electronic device **111** may, in accordance with the recognition, recognize, determine, or identify ceasing the first audio service being provided using the electronic device **101** and providing the second audio service by using the electronic device **101**. For example, the first external electronic device **111** may transmit, to the electronic device **101**, the request, based on a determination that ceases the first audio service being provided using the electronic device **101** and provides the second audio service by using the electronic device **101**.

[0099] As a non-limiting example, the request may be transmitted before operations for providing the second audio service are executed.

[0100] As a non-limiting example, the request may be transmitted through a modify source operation of a broadcast audio scan service (BASS).

[0101] For example, the request may be indicated via a PA_Sync field and a BIS_Sync field for the modify source operation. For example, for the request that maintains the synchronizing to the periodic advertisement **142**, the PA_Sync field may be set, among a first value (e.g., ‘0×01’ (synchronize to PA-PAST available)), a second value (e.g., ‘0×02’ (synchronize to PA-PAST not available)), and a third value (e.g., ‘0×00’ (do not synchronize to PA)), to the first value or the second value. For example, for the request that ceases the synchronizing to the BIS, all BIS subgroup indexes in the BIS_Sync field may be set, among a fourth value (e.g. ‘0b0’ (Do not synchronize to BIS_index[x]), wherein x indicates the number of the index) and a fifth value (e.g. ‘0b1’ (synchronize to BIS_index[x]), to the fourth value. For example, for the request that maintains the synchronizing to the BIS, the BIS_Sync field may be set to the fifth value among the fourth value and the fifth value. When the BIS_Sync field is set to the fifth value for the request, a metadata field for the request may indicate ceasing rendering of the data for the BIS.
[0102] For another example, the request may be indicated via a metadata field for the modify source operation. For example, the metadata field may indicate maintaining the synchronizing to the periodic advertisement **142** and ceasing the synchronizing to the BIS. For another example, the metadata field may indicate maintaining the synchronizing to the periodic advertisement **142**, maintaining the synchronizing to the BIS, and ceasing rendering of the data for the BIS.
[0103] As a non-limiting example, the request may be transmitted via a remove source operation and an add source operation of the BASS. For example, a metadata field for the add source operation may indicate maintaining the synchronizing to the periodic advertisement **142** and ceasing the synchronizing to the BIS, unlike the remove source operation indicating a request that ceases the synchronizing to the periodic advertisement **142** and ceases the synchronizing to the BIS. For another example, the metadata field for the add source operation may indicate maintaining the synchronizing to the periodic advertisement **142**, maintaining the synchronizing to the BIS, and ceasing rendering of the data for the BIS, unlike the remove source operation indicating a request that ceases synchronizing to the periodic advertisement **142** and ceases the synchronizing to the BIS.
[0104] Referring back to FIG. **4**, the electronic device **101** may receive, from the first external electronic device **111** via the communication link **131**, the request, in accordance with operation **408**.
[0105] In operation **409**, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** based on the request, while providing the second audio service in conjunction with the first external electronic device **111**.
[0106] Although not illustrated in FIG. **4**, as a non-limiting example, the first external electronic device **111** may, after transmitting the request (or along with transmitting the request), execute operations to start providing the second audio service.
[0107] For example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal (e.g., including call start information) in accordance with a hands-free profile (HFP), for starting the second audio service, based on detecting the event exemplified in the description of operation **602**. For example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal (e.g., including call start information) in accordance with a telephone bearer service (TBS), for starting the second audio service, based on detecting the event. For example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal (e.g., including media playback start information) in accordance with an advanced audio distribution profile (A2DP). For example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal (e.g., including media playback start information) in accordance with a media control profile (MCP).
[0108] Although not illustrated in FIG. **4**, as a non-limiting example, the first external electronic device **111** may execute operations to provide the second audio service.

[0109] For example, the first external electronic device **111** may connect or establish an audio channel with the electronic device **101** via the communication link **131**, and transmit data for the second audio service to the electronic device **101** via the connected audio channel. For example, the first external electronic device **111** may receive, from the electronic device **101** through the audio channel, data for the second audio service. For example, the second audio service may be provided according to outputting audio via the speaker **303** of the electronic device **101** based on data transmitted from the first external electronic device **111** via the audio channel. For example, the second audio service may be provided according to transmitting data for audio obtained via the microphone **304** of the electronic device **101** to the first external electronic device **111** via the audio channel.

[0110] As a non-limiting example, the first external electronic device **111** may connect or establish a link of synchronous connection-oriented (SCO) via the communication link **131**, and provide the second audio service by transmitting or receiving the data for the second audio service via the link of the SCO. As a non-limiting example, the first external electronic device **111** may connect or establish a link of an extended SCO (eSCO) via the communication link **131**, and provide the second audio service by transmitting or receiving the data for the second audio service via the link of the eSCO. As a non-limiting example, the first external electronic device **111** may connect or establish a link of a connected isochronous stream (CIS) via the communication link **131**, and provide the second audio service by transmitting or receiving the data for the second audio service through the link of the CIS.

[0111] For example, the electronic device **101** may execute a first handover from the first audio service to the second audio service, based on connecting the audio channel exemplified above, in conjunction with the first external electronic device **111**. For example, the electronic device **101** may provide, based on the first handover, the second audio service in conjunction with the first external electronic device **111**. For example, the second audio service provided in conjunction with the first external electronic device **111** may include outputting audio via the speaker **303** of the electronic device **101** based on data received from the first external electronic device **111**. For example, the second audio service provided in conjunction with the first external electronic device **111** may include transmitting information on audio obtained via the microphone **304** of the electronic device **101** to the first external electronic device **111**.

[0112] For example, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** based on the request, while providing the second audio service based on data received from the first external electronic device **111** via the communication link **131**. For example, for the first audio service capable of being resumed in accordance with the second handover from the second audio service to the first audio service, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142**. For example, in order to execute resuming the first audio service with a faster response speed, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** while the second audio service is provided.

[0113] For example, maintaining the synchronizing to the periodic advertisement **142** based on the request while the second audio service is provided may include receiving at least a portion of packets advertised from the second external electronic device **112** in accordance with the periodic advertisement **142** while the second audio service is provided. As a non-limiting example, maintaining the synchronizing to the periodic advertisement **142** based on the request while the second audio service is provided may include receiving all of the packets advertised from the second external electronic device **112** in accordance with the periodic advertisement **142** while the second audio service is provided. As a non-limiting example, maintaining the synchronizing to the periodic advertisement **142** based on the request while the second audio service is provided may include receiving a portion of the packets advertised from the second external electronic device **112** in accordance with the periodic advertisement **142** while the second audio service is provided. For example, the electronic device **101** may receive, while the second audio service is provided, at least one packet advertised from the second external electronic device **112** via the periodic advertisement **142** of an event of the 2n-1 counter, and bypass, while the second audio service is provided, receiving at least one packet advertised from the second external electronic device **112** via the periodic advertisement **142** of an event of the 2n counter. 'n' may be any natural number greater than or equal to 1. As a non-limiting example, specifying packets to be received by the electronic device **101** according to maintaining the periodic advertisement **142** may be set by the electronic device **101** or may be set by the first external electronic device **111**. For example, when the packets are specified by the first external electronic device **111**, the setting of the first external electronic device **111** may be indicated via a metadata field for the modify source operation. However, it is not limited thereto.

[0114] For example, the electronic device **101** may cease, based on the request indicating ceasing the synchronizing to the BIS from the second external electronic device **112**, the synchronizing to the BIS while the second audio service is provided. For example, the electronic device **101** may, based on the request indicating maintaining the synchronizing to the BIS from the second external electronic device **112**, maintain the synchronizing to the BIS while the second audio service is provided. The ceasing the synchronizing to the BIS and the maintaining the synchronizing to the BIS are exemplified in a description of FIG. **7** and a description of FIG. **8**.

[0115] FIGS. **7** and **8** illustrate a method executed in an electronic device in accordance with reception of a request according to various embodiments of the disclosure.

[0116] Referring to FIG. **7**, in operation **701**, the electronic device **101** may receive, from the first external electronic device **111** via the communication link **131**, the request. Operation **701** may correspond to operation **408** of FIG. **4**.

[0117] In operation **702**, the electronic device **101** may cease, based on the request, providing the first audio service by ceasing the synchronizing to the BIS from the second external electronic device **112**, while the second audio service is being provided. For example, on a condition that the providing of the second audio service is started within the electronic device **101**, the electronic device **101** may cease, terminate, refrain from, bypass, or skip receiving the data for the BIS from the second external electronic device **112**, based on the synchronizing to the periodic advertisement **142** maintained in accordance with the request. For example, since the receiving of the data for the BIS is ceased in accordance with the request, the first audio service may be ceased.

[0118] Referring to FIG. **8**, in operation **801**, the electronic device **101** may receive, from the first external electronic device **111** via the communication link **131**, the request. Operation **801** may correspond to operation **408** of FIG. **4**.

[0119] In operation **802**, the electronic device **101** may cease, based on the request, providing the first audio service, while the second audio service is being provided, by ceasing rendering the data for the BIS broadcasted from the second external electronic device **112** in accordance with the synchronizing to the BIS from the second external electronic device **112**. For example, the electronic device **101** may maintain, based on the request, receiving the data for the BIS from the second external electronic device **112**, in accordance with the synchronizing to the periodic advertisement **142** being maintained via the request. For example, on a condition that the providing of the second audio service is started within the electronic device **101**, the electronic device **101** may, based on the request, cease, terminate, refrain from, bypass, or skip rendering the data for the BIS received from the second external electronic device **112**. For example, since the rendering of the data for the BIS is ceased in accordance with the request, the first audio service may be ceased.

[0120] As described above, the electronic device **101** may maintain, while the second audio service is provided, the synchronizing to the periodic advertisement **142** for the first audio service, based on the request received from the first external electronic device **111**. As exemplified in the description of FIG. **9** below, the electronic device **101** may resume providing the first audio service with a faster speed in response to another event for terminating the second audio service, by maintaining the synchronizing to the periodic advertisement **142** while the second audio service is provided. Resuming the first audio service is exemplified in a description of FIG. **9**. The description of FIGS. **4** to **8** describes operations executed within the electronic device **101** and the first external electronic device **111** when an audio service provided using the electronic device **101** is changed from the first audio service to the second audio service, but this is only an example. The operations may also be executed to change the first audio service to a third audio service provided based on a broadcast from another external electronic device (hereinafter, a third external electronic device).

[0121] For example, the first external electronic device **111** may receive a user input indicating a request to provide the third audio service using the electronic device **101**, while providing the first audio service, as in operation **407**. For example, the first external electronic device **111** may, in response to the user input, transmit a request to the electronic device **101** via the communication link **131**, as in operation **408**. The request may include information on a periodic advertisement of the third external electronic device. The request may indicate maintaining the synchronizing to the BIS from the second external electronic device **112**, as in the request in operation **408**. For example, in response to receiving the request from the first external electronic device **111**, the electronic device **101** may execute, based on the information on the periodic advertisement of the third external electronic device, synchronizing to the periodic advertisement of the third external electronic device for the third audio service, and execute, based on the synchronizing to the periodic advertisement of the third external electronic device, synchronizing to the BIS from the third external electronic device. For example, the electronic device **101** may provide, based on the synchronizing to the BIS from the third external electronic device, the third audio service. For example, in response to receiving the request from the first external electronic device **111**, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** while executing at least one operation for the third audio service (e.g., executing the synchronizing to the periodic advertisement of the third external electronic device, executing the synchronizing to the BIS from the third external electronic device, and/or providing the third audio service). For example, maintaining the synchronizing to the periodic advertisement **142** may be executed for responsiveness of a change from the third audio service to the first audio service.

[0122] As another example, the electronic device **101** may detect an event (or a user input (e.g., a press input on a physical button exposed through a portion of a housing of the electronic device **101**) for starting the third audio service, while providing the first audio service. For example, in response to the event, the electronic device **101** may attempt synchronizing to the periodic advertisement of the third external electronic device for the third audio service. As a non-limiting example, the electronic device **101** may execute the synchronizing to the periodic advertisement of the third external electronic device, based on information on the periodic advertisement of the third external electronic device received from the first external electronic device **111**. For example, the electronic device **101** may execute synchronizing to the BIS from the third external electronic device, based on the synchronizing of the third external electronic device to the periodic advertisement. For example, the electronic device **101** may provide the third audio service, based on the synchronizing to the BIS from the third external electronic device. For example, providing the first audio service may be ceased in accordance with providing the third audio service. For example, in response to the user input, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142**, while executing at least one operation (e.g., executing the synchronizing to the periodic advertisement of the third external electronic device, executing the synchronizing to the BIS from the third external electronic device, and/or providing the third audio service) for the third audio service. For example, maintaining the synchronizing to the periodic advertisement **142** may be executed for responsiveness of a change from the third audio service to the first audio service.

[0123] FIG. **9** illustrates a method of resuming synchronizing to a BIS based on another request caused in accordance with a second handover of an audio service according to an embodiment of

the disclosure.
[0124] Referring to FIG. **9**, in operation **901**, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142**, while providing the second audio service, in conjunction with the first external electronic device **111**. For example, operation **901** may correspond to operation **409** of FIG. **4**.
[0125] In operation **903**, the second external electronic device **112** may execute the periodic advertisement **142**. For example, since the first audio service provided from the second external electronic device **112** may not be a service specified or dedicated (only) for the electronic device **101**, the second external electronic device **112** may execute the periodic advertisement **142** regardless of the first audio service ceased within the electronic device **101**. For example, the electronic device **101** may maintain, while the second audio service is provided, the synchronizing to the periodic advertisement **142** executed by the second external electronic device **112** in operation **903**, as in operation **901**. For example, operation **903** may correspond to operation **403** of FIG. **4**.
[0126] In operation **904**, the second external electronic device **112** may execute broadcasting (**143**) the data for the BIS from the second external electronic device **112** associated with the periodic advertisement **142**, as indicated by arrow **152**. For example, since the first audio service provided from the second external electronic device **112** may not be a service specified or dedicated (only) for the electronic device **101**, the second external electronic device **112** may maintain broadcasting the data for the BIS, regardless of the first audio service ceased within the electronic device **101**.
[0127] In operation **902**, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, another request to resume the synchronizing to the BIS.
[0128] For example, the other request may be transmitted in accordance with detection of another event for termination (or cessation) of the second audio service. The other request transmitted in response to the other event is exemplified in a description of FIG. **10**.
[0129] FIG. **10** illustrates a method executed in a first external electronic device in accordance with detection of another event according to an embodiment of the disclosure.
[0130] Referring to FIG. **10**, in operation **1001**, the first external electronic device **111** may detect another event for ceasing the second audio service.
[0131] For example, in response to the other event, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, information to terminate the second audio service. As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal in accordance with HFP (e.g., including call termination information) (not illustrated in FIG. **10**). As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal in accordance with TBS (e.g., including call termination information) (not illustrated in FIG. **10**). As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal in accordance with A2DP (e.g., including media playback termination information) (not illustrated in FIG. **10**). As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal in accordance with MCP (e.g., including media playback termination information) (not illustrated in FIG. **10**).
[0132] As a non-limiting example, the first external electronic device **111** may disconnect or release an audio channel (e.g., a link of SCO, a link of eSCO, or a link of CIS) for the second audio service after transmitting the information or based on transmitting the information.
[0133] In operation **1002**, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, the other request exemplified in the description of operation **902** of FIG. **9**, based on the other event.
[0134] For example, the other request may be transmitted via a modify source operation of BASS.

[0135] For example, on a condition that the request transmitted to the electronic device **101** in operation **408** is a request that ceases the synchronizing to the BIS, the other request may be indicated via a BIS_Sync field for the modify source operation. For example, at least a portion of BIS subgroup indexes of the BIS_Sync field may be set to the fifth value among the fourth value and the fifth value, for the other request to resume the synchronizing to the BIS.
[0136] For example, on a condition that the request transmitted to the electronic device **101** in operation **408** is a request that ceases the synchronizing to the BIS, or a request that maintains the synchronizing to the BIS and ceases rendering of the data for the BIS, the other request may indicate via a metadata field for the modify source. For example, the metadata field may indicate resuming the synchronizing to the BIS on a condition that the request transmitted to the electronic device **101** in operation **408** is a request that ceases the synchronizing to the BIS. For example, the metadata field may indicate resuming rendering of the data for the BIS on a condition that the request transmitted to electronic device **101** in operation **408** is a request that maintains the synchronizing to the BIS and ceases rendering of the data for the BIS.
[0137] For example, on a condition that the request transmitted to the electronic device **101** in operation **408** was transmitted via a remove source operation and an add source operation, the other request may be transmitted via an add source operation of BASS. For example, the other request may be indicated via a BIS_Sync field for the add source operation. For example, at least a portion of BIS subgroup indexes of the BIS_Sync field may be set to the fifth value among the fourth value and the fifth value, for the other request to resume the synchronizing to the BIS.
[0138] Referring back to FIG. **9**, the electronic device **101** may receive, from the first external electronic device **111** via the communication link **131**, the other request.
[0139] In operation **905**, in response to the other request, the electronic device **101** may resume the first audio service by resuming the synchronizing to the BIS from the second external electronic device **112** as indicated by arrow **161**, in accordance with the synchronizing to the periodic advertisement **142** maintained while the second audio service is provided. For example, in response to the other request, the electronic device **101** may resume the first audio service based on the second handover by resuming the synchronizing to the BIS. For example, the electronic device **101** may cease the second audio service and resume the first audio service, based at least in part on the other request. For example, the electronic device **101** may ceases the second audio service to resume the first audio service, in response to the other request.
[0140] The operations exemplified in the descriptions of FIGS. **4** to **10** may be executed to maintain the periodic advertisement **142** on a condition that the first external electronic device **111** is a device capable of transmitting the request **408** and the other request **902**. However, this is merely an example. For example, the electronic device **101** may maintain the periodic advertisement **142** for the first audio service while the second audio service is provided, via communication with a device incapable of transmitting the request **408** and the other request **902**, rather than a device capable of transmitting the request **408** and the other request **902** (e.g., the first external electronic device **111**). The operations of the electronic device **101** for maintaining the periodic advertisement **142** for the first audio service while the second audio service is provided without the request **408** and the other request **902** are exemplified in a description of FIG. **11**. The first external electronic device **111** exemplified in accordance with the description of FIG. **11** may be a device that does not transmit the request **408** and the other request **902**.
[0141] FIG. **11** illustrates a method of maintaining a periodic advertisement based on a signal caused for a first handover of an audio service according to an embodiment of the disclosure.
[0142] Referring to FIG. **11**, in operation **1101**, the electronic device **101** and the first external electronic device **111** may establish a communication link **131**. For example, operation **1101** may correspond to operation **401** of FIG. **4**.
[0143] In operation **1102**, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, information on a periodic advertisement **142** of the

second external electronic device **112**. For example, operation **1102** may correspond to operation **402** of FIG. **4**.

[0144] In operation **1103**, the second external electronic device **112** may execute the periodic advertisement **142**. For example, operation **1103** may correspond to operation **403** of FIG. **4**.

[0145] In operation **1104**, the second external electronic device **112** may execute broadcasting (**143**) data for the BIS indicated by the periodic advertisement **142**. For example, operation **1104** may correspond to operation **404** of FIG. **4**.

[0146] In operation **1105**, the electronic device **101** may synchronize to the periodic advertisement **142**. For example, operation **1105** may correspond to operation **405** of FIG. **4**.

[0147] In operation **1106**, the electronic device **101**) may synchronize to the BIS from the second external electronic device **112**, as indicated by arrow **161**, in accordance with the synchronizing to the periodic advertisement **142** executed in operation **1105**. For example, operation **1106** may correspond to operation **406** of FIG. **4**.

[0148] In operation **1107**, the electronic device **101** may provide the first audio service outputting audio via the speaker **303**, based on the synchronizing to the BIS in operation **1106**. Operation **1107** may correspond to operation **407** of FIG. **4**.

[0149] In operation **1108**, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, a signal for a start of the second audio service via the communication link **131**, while the first audio service is provided. For example, the signal may indicate requesting operations for starting the second audio service.

[0150] For example, the signal may be variously implemented. As a non-limiting example, the signal may include call start information in accordance with HFP. As a non-limiting example, the signal may include call start information in accordance with TBS. As a non-limiting example, the signal may include media playback start information in accordance with A2DP. As a non-limiting example, the signal may include media playback start information in accordance with MCP.

[0151] For example, the electronic device **101** may receive, from the first external electronic device **111** via the communication link **131**, the signal.

[0152] In operation **1109**, based on the signal, the electronic device **101** may maintain synchronizing to the periodic advertisement **142**, while ceasing providing the first audio service and providing the second audio service in conjunction with the first external electronic device **111**.

[0153] As a non-limiting example, in response to the signal, the electronic device **101** may compare a priority of the first audio service with a priority of the second audio service. For example, based on the comparison, the electronic device **101** may recognize, determine, or check that a priority of the first audio service provided via the second external electronic device **112** with which a communication link is not established is lower than a priority of the second audio service provided via the first external electronic device **111** with which a communication link **131** is established. For example, based on the comparison, the electronic device **101** may recognize, determine, or check that a priority of the first audio service provided based on connectionless is lower than a priority of the second audio service provided based on a connection.

[0154] As a non-limiting example, in response to the signal, the electronic device **101** may compare a priority of the first external electronic device **111** with a priority of the second external electronic device **112**. For example, based on the comparison, the electronic device **101** may determine, recognize, or check that a priority of the first external electronic device **111** that has establish the communication link **131** is higher than a priority of the second external electronic device **112** that has not establish a communication link. For example, based on the comparison, the electronic device **101** may determine, recognize, or check that a priority of the first external electronic device **111** that is available for controlling the electronic device **101** is higher than a priority of the second external electronic device **112** that is unavailable for controlling the electronic device **101**. For example, based on the comparison, the electronic device **101** may determine, recognize, or check that a priority of the first external electronic device **111** that is logged in a user account corresponding to a user account of the electronic device **101** is higher than a priority of the second external electronic device **112** that is logged in a user account different from the user account of the electronic device **101**.

[0155] As a non-limiting example, based on a result of the comparison, the electronic device **101** may determine to cease providing the first audio service and provide the second audio service, in response to the signal.

[0156] As a non-limiting example, providing the first audio service may be variously implemented. For example, the electronic device **101** may cease providing the first audio service by ceasing the synchronizing to the BIS in response to the signal. For example, the electronic device **101** may cease providing the first audio service, by ceasing rendering of the data for the BIS received from the second external electronic device **112** in accordance with the synchronizing to the BIS in response to the signal.

[0157] Although not illustrated in FIG. **11**, as a non-limiting example, the first external electronic device **111** may execute operations for providing the second audio service.

[0158] For example, the first external electronic device **111** may connect or establish, via the communication link **131**, an audio channel with the electronic device **101**, and transmit, to the electronic device **101** via the connected audio channel, data for the second audio service. For example, the first external electronic device **111** may receive, from the electronic device **101** via the audio channel, data for the second audio service. For example, the second audio service may be provided in accordance with outputting audio via the speaker **303** of the electronic device **101** based on data transmitted from the first external electronic device **111** via the audio channel. For example, the second audio service may be provided in accordance with transmitting data on audio obtained via the microphone **304** of the electronic device **101** to the first external electronic device **111** via the audio channel.

[0159] As a non-limiting example, the first external electronic device **111** may connect or establish a link of a synchronous connection-oriented (SCO) via the communication link **131**, and provide the second audio service by transmitting or receiving the data for the second audio service via the link of the SCO. As a non-limiting example, the first external electronic device **111** may connect or establish a link of extended SCO (eSCO) via the communication link **131**, and provide the second audio service by transmitting or receiving the data for the second audio service via the link of the eSCO. As a non-limiting example, the first external electronic device **111** may connect or establish a link of a connected isochronous stream (CIS) via the communication link **131**, and provide the second audio service by transmitting or receiving the data for the second audio service via the link of the CIS.

[0160] For example, the electronic device **101** may execute the first handover from the first audio service to the second audio service, based on connecting the audio channel exemplified above, in conjunction with the first external electronic device **111**. For example, based on the first handover, the electronic device **101** may provide the second audio service in conjunction with the first external electronic device **111**. For example, the second audio service provided in conjunction with the first external electronic device **111** may include outputting audio via the speaker **303** of the electronic device **101** based on data received from the first external electronic device **111**. For example, the second audio service provided in conjunction with the first external electronic device **111** may include transmitting, to the first external electronic device **111**, information on audio obtained via the microphone **304** of the electronic device **101**.

[0161] For example, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** based on the signal, while providing the second audio service based on data received from the first external electronic device **111** via the communication link **131**. For example, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** for the first audio service capable of being resumed in accordance with a second handover from the second audio service to the first audio service. For example, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** while the second audio service is provided, in order to execute resuming of the first audio service with a faster response speed. For example, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142** even when the first audio service is ceased, to resume the first audio service with a faster response in accordance with the interruption of the second audio service capable of being caused later.

[0162] As described above, the electronic device **101** may maintain, while the second audio service is provided, the synchronizing to the periodic advertisement **142** for the first audio service, based on the signal received from the first external electronic device **111**. The electronic device **101** may resume providing the first audio service with a faster speed in response to another event for terminating the second audio service, via maintaining the synchronizing to the periodic advertisement **142** while the second audio service is provided, as exemplified in the description of FIG. **12** below. Resuming the first audio service is exemplified in the description of FIG. **12**.

[0163] FIG. **12** illustrates a method of resuming synchronizing to a BIS based on another signal caused for a second handover of an audio service according to an embodiment of the disclosure.

[0164] Referring to FIG. **12**, in operation **1201**, the electronic device **101** may maintain the synchronizing to the periodic advertisement **142**, while providing the second audio service, in conjunction with the first external electronic device **111**. For example, operation **1201** may correspond to operation **901** of FIG. **9**.

[0165] In operation **1203**, the second external electronic device **112** may execute the periodic advertisement **142**. For example, operation **1203** may correspond to operation **903** of FIG. **9**.

[0166] In operation **1204**, the second external electronic device **112** may broadcast the data for the BIS from the second external electronic device **112** associated with the periodic advertisement **142**, as indicated by arrow **152**. For example, operation **1204** may correspond to operation **904** of FIG. **9**.

[0167] In operation **1202**, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, another signal for terminating the second audio service.

[0168] For example, the other signal may be variously implemented. As a non-limiting example, the other signal may include call termination information in accordance with HFP. As a non-limiting example, the other signal may include call termination information in accordance with TBS. As a non-limiting example, the other signal may include media playback termination information in accordance with A2DP. As a non-limiting example, the other signal may include media playback termination information in accordance with MCP.

[0169] For example, the other signal may be transmitted in response to detecting another event for terminating the second audio service.

[0170] For example, in response to the other event, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, information for terminating the second audio service. As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, the other signal in accordance with HFP (e.g., including call termination information). As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, the other signal in accordance with TBS (e.g., including call termination information). As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, the other signal in accordance with A2DP (e.g., including media playback termination information). As a non-limiting example, the first external electronic device **111** may transmit, to the electronic device **101** via the communication link **131**, the other signal in accordance with MCP (e.g., including media playback termination information).

[0171] As a non-limiting example, the first external electronic device **111** may disconnect or release an audio channel (e.g., a link of SCO, a link of eSCO, or a link of CIS) for the second audio service, after transmitting the other signal or based on transmitting the other signal.

[0172] For example, the electronic device **101** may receive, from the first external electronic device **111** via the communication link **131**, the other signal.

[0173] In operation **1205**, based on the other signal, the electronic device **101** may resume the first audio service by resuming synchronizing to the BIS, as indicated by arrow **161**, in accordance with synchronizing to the periodic advertisement **142** maintained while the second audio service is provided. For example, based on the other signal, the electronic device **101** may resume the first audio service based on the second handover by resuming the synchronizing to the BIS. For example, the electronic device **101** may cease the second audio service and resume the first audio service, based at least in part on the other signal.

[0174] FIG. **12** illustrates resuming the synchronizing to the BIS to resume the first audio service, but this is merely an example. For example, the electronic device **101** may resume the first audio service by resuming rendering of the data for the BIS, on a condition of maintaining the synchronizing to the BIS and ceasing rendering of the data for the BIS in operation **1109**.

[0175] As described above, the electronic device **101** may, in response to the other signal, resume the first audio service in accordance with synchronizing to the periodic advertisement **142** that was maintained based on the signal (e.g., operation **1108**). For example, the electronic device **101** may improve quality of an audio service through this operation.

[0176] The operations of the electronic device **101** and the first external electronic device **111** exemplified above may be executed by the electronic device **1301** and the electronic device **1302** in the description of FIG. **13** described below.

[0177] FIG. **13** is a block diagram illustrating an electronic device **1301** in a network environment **1300** according to an embodiment of the disclosure. Referring to FIG. **13**, the electronic device **1301** in the network environment **1300** may communicate with an electronic device **1302** via a first network **1398** (e.g., a short-range wireless communication network), or at least one of an electronic device **1304** or a server **1308** via a second network **1399** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **1301** may communicate with the electronic device **1304** via the server **1308**. According to an embodiment, the electronic device **1301** may include a processor **1320**, memory **1330**, an input module **1350**, a sound output module **1355**, a display module **1360**, an audio module **1370**, a sensor module **1376**, an interface **1377**, a connecting terminal **1378**, a haptic module **1379**, a camera module **1380**, a power management module **1388**, a battery **1389**, a communication module **1390**, a subscriber identification module (SIM) **1396**, or an antenna module **1397**. In some embodiments, at least one of the components (e.g., the connecting terminal **1378**) may be omitted from the electronic device **1301**, or one or more other components may be added in the electronic device **1301**. In some embodiments, some of the components (e.g., the sensor module **1376**, the camera module **1380**, or the antenna module **1397**) may be implemented as a single component (e.g., the display module **1360**).

[0178] The processor **1320** may execute, for example, software (e.g., a program **1340**) to control at least one other component (e.g., a hardware or software component) of the electronic device **1301** coupled with the processor **1320**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **1320** may store a command or data received from another component (e.g., the sensor module **1376** or the communication module **1390**) in volatile memory **1332**, process the command or the data stored in the volatile memory **1332**, and store resulting data in non-volatile memory **1334**. According to an embodiment, the processor **1320** may include a main processor **1321** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **1323** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **1321**. For example, when the electronic device

**1301** includes the main processor **1321** and the auxiliary processor **1323**, the auxiliary processor **1323** may be adapted to consume less power than the main processor **1321**, or to be specific to a specified function. The auxiliary processor **1323** may be implemented as separate from, or as part of the main processor **1321**.
[0179] The auxiliary processor **1323** may control at least some of functions or states related to at least one component (e.g., the display module **1360**, the sensor module **1376**, or the communication module **1390**) among the components of the electronic device **1301**, instead of the main processor **1321** while the main processor **1321** is in an inactive (e.g., sleep) state, or together with the main processor **1321** while the main processor **1321** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **1323** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **1380** or the communication module **1390**) functionally related to the auxiliary processor **1323**. According to an embodiment, the auxiliary processor **1323** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **1301** where the artificial intelligence is performed or via a separate server (e.g., the server **1308**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.
[0180] The memory **1330** may store various data used by at least one component (e.g., the processor **1320** or the sensor module **1376**) of the electronic device **1301**. The various data may include, for example, software (e.g., the program **1340**) and input data or output data for a command related thereto. The memory **1330** may include the volatile memory **1332** or the non-volatile memory **1334**.
[0181] The program **1340** may be stored in the memory **1330** as software, and may include, for example, an operating system (OS) **1342**, middleware **1344**, or an application **1346**.
[0182] The input module **1350** may receive a command or data to be used by another component (e.g., the processor **1320**) of the electronic device **1301**, from the outside (e.g., a user) of the electronic device **1301**. The input module **1350** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
[0183] The sound output module **1355** may output sound signals to the outside of the electronic device **1301**. The sound output module **1355** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.
[0184] The display module **1360** may visually provide information to the outside (e.g., a user) of the electronic device **1301**. The display module **1360** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **1360** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.
[0185] The audio module **1370** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **1370** may obtain the sound via the input module **1350**, or output the sound via the sound output module **1355** or a headphone of an external electronic device (e.g., an electronic device **1302**) directly (e.g., wiredly) or wirelessly coupled

with the electronic device **1301**.

[0186] The sensor module **1376** may detect an operational state (e.g., power or temperature) of the electronic device **1301** or an environmental state (e.g., a state of a user) external to the electronic device **1301**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **1376** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0187] The interface **1377** may support one or more specified protocols to be used for the electronic device **1301** to be coupled with the external electronic device (e.g., the electronic device **1302**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **1377** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0188] A connecting terminal **1378** may include a connector via which the electronic device **1301** may be physically connected with the external electronic device (e.g., the electronic device **1302**). According to an embodiment, the connecting terminal **1378** may include, for example, an HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0189] The haptic module **1379** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **1379** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0190] The camera module **1380** may capture a still image or moving images. According to an embodiment, the camera module **1380** may include one or more lenses, image sensors, image signal processors, or flashes.

[0191] The power management module **1388** may manage power supplied to the electronic device **1301**. According to an embodiment, the power management module **1388** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0192] The battery **1389** may supply power to at least one component of the electronic device **1301**. According to an embodiment, the battery **1389** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0193] The communication module **1390** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **1301** and the external electronic device (e.g., the electronic device **1302**, the electronic device **1304**, or the server **1308**) and performing communication via the established communication channel. The communication module **1390** may include one or more communication processors that are operable independently from the processor **1320** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **1390** may include a wireless communication module **1392** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **1394** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **1398** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **1399** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **1392** may identify and authenticate the electronic device **1301** in a communication network, such as the first network **1398** or the second network **1399**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **1396**.

[0194] The wireless communication module **1392** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **1392** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **1392** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **1392** may support various requirements specified in the electronic device **1301**, an external electronic device (e.g., the electronic device **1304**), or a network system (e.g., the second network **1399**). According to an embodiment, the wireless communication module **1392** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 1364 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 13 ms or less) for implementing URLLC.

[0195] The antenna module **1397** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **1301**. According to an embodiment, the antenna module **1397** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **1397** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **1398** or the second network **1399**, may be selected, for example, by the communication module **1390** (e.g., the wireless communication module **1392**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **1390** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **1397**.

[0196] According to various embodiments, the antenna module **1397** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0197] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0198] According to an embodiment, commands or data may be transmitted or received between the electronic device **1301** and the external electronic device **1304** via the server **1308** coupled with the second network **1399**. Each of the electronic devices **1302** or **1304** may be a device of a same type as, or a different type, from the electronic device **1301**. According to an embodiment, all or some of operations to be executed at the electronic device **1301** may be executed at one or more of the external electronic devices **1302** or **1304**, or the server **1308**. For example, if the electronic device **1301** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **1301**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **1301**. The electronic device **1301** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **1301** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **1304** may include an internet-of-things (IoT) device. The server **1308** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **1304** or the server **1308** may be included in the second network **1399**. The electronic device **1301** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0199] As described above, an electronic device (e.g., the electronic device **101**) may comprise memory configured to store instructions, communication circuitry (e.g., the communication circuitry **302**) for Bluetooth low energy (BLE), a speaker (e.g., the speaker **303**), and at least one processor (e.g., the at least one processor **301**). The at least one processor may be configured to execute the instructions to cause the electronic device to, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device (e.g., the second external electronic device **112**) indicated by information from a first external electronic device (e.g., the first external electronic device **111**), to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via the speaker. The at least one processor may be configured to execute the instructions to cause the electronic device to, while the first audio service is provided, receive, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a request that ceases the synchronizing to the BIS and maintains the synchronizing to the periodic advertisement, using the communication circuitry. The at least one processor may be configured to execute the instructions to cause the electronic device to, while providing a second audio service based on data received via the communication link from the first external electronic device, cease the synchronizing to the BIS and maintain the synchronizing to the periodic advertisement.

[0200] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to maintain the synchronizing to the periodic advertisement by receiving at least a portion of packets advertised from the second external electronic device in accordance with the periodic advertisement based on the request while providing the second audio service.

[0201] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to receive the request from the first external electronic device, through a modify source operation of a broadcast audio scan service (BASS).

[0202] For example, the request may indicate maintaining the synchronizing to the periodic advertisement using a PA_sync filed of the modify source operation, and ceasing the synchronizing to the BIS, using a BIS_sync field of the modify source operation.

[0203] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to, while providing the second audio service, receive, via the communication link, from the first external electronic device, another request to resume the synchronizing to the BIS, using the communication circuitry, and in response to the other request, resume to provide the first audio service by resuming the synchronizing to the BIS in accordance with the synchronizing to the periodic advertisement maintained based on the request.

[0204] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to, through a modify source operation of BASS, receive the other request from the first external electronic device.

[0205] For example, the other request may indicate resuming the synchronizing to the BIS, using a BIS_sync filed of the modify source operation.

[0206] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to cease providing the second audio service, for resuming to provide the first audio service in response to the other request.

[0207] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to, based on outputting audio via the speaker based on data received via the communication link from the first external electronic device, provide the second audio service.

[0208] For example, the electronic device may comprise a microphone (e.g., the microphone **304**). For example, the at least one processor may be configured to execute the instructions to cause the electronic device to, based on transmitting, via the communication link, to the first external electronic device, data regarding audio obtained via the microphone using the communication circuitry, provide the second audio service.

[0209] As described above, an electronic device (e.g., the electronic device **101**) may comprise memory configured to store instructions, communication circuitry (e.g., the communication circuitry **302**) for Bluetooth low energy (BLE), a speaker (e.g., the speaker **303**), and at least one processor (e.g., the at least one processor **301**). The at least one processor may be configured to execute the instructions to cause the electronic device to, based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device (e.g., the second external electronic device **112**) indicated by information from a first external electronic device (e.g., the first external electronic device **111**), to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via the speaker, while the first audio service is provided, receive, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a signal for starting a second audio service, using the communication circuitry, in response to the signal, cease providing the first audio service based on data for the BIS broadcasted from the second external electronic device, and based on the signal, maintain the synchronizing to the periodic advertisement, while ceasing providing the first audio service and providing the second audio service based on data received via the communication link from the first external electronic device.

[0210] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to maintain the synchronizing of the periodic advertisement, by receiving at least a portion of packets advertised from the second external electronic device in accordance with the periodic advertisement, based on the signal, while providing the second audio service.

[0211] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to cease providing the first audio service, by ceasing receiving the data for the BIS, in response to the signal.

[0212] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to cease providing the first audio service, by ceasing rendering the received data for the BIS, in accordance with the synchronizing to the periodic advertisement from the second external electronic device, using the communication circuitry, in response to the signal.

[0213] For example, the signal may include call start information in accordance with a hands-free profile (HFP) or a telephone bearer service (TBS).

[0214] For example, the signal may include media playback start information in accordance with an advanced audio distribution profile (A2DP) or a media control profile (MCP).

[0215] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to receive, from the first external electronic device via the communication link, another signal for terminating the second audio service using the communication circuitry, while providing the second audio service, and resume providing the first audio service, by resuming the synchronizing to the BIS in accordance with the synchronizing to the periodic advertisement maintained based on the signal in response to the other signal.

[0216] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to transmit, to the first external electronic device via the communication link, information indicating that the electronic device is synchronized to the BIS, using the communication circuitry.

[0217] As described above, an electronic device (e.g., the first external electronic device **111**) may comprise communication circuitry (e.g., the communication circuitry **202**) for Bluetooth low energy (BLE) and at least one processor (e.g., the at least one processor **201**). For example, the at least one processor may be configured to execute the instructions to cause the electronic device to transmit, to another electronic device (e.g., the electronic device **101**) via a communication link between the electronic device and the other electronic device, first information on a periodic advertisement of an external electronic device (e.g., the second external electronic device **112**), using the communication circuitry, receive, via the communication link, from the other electronic device, second information indicating a state of the other electronic device synchronized to a broadcast isochronous stream (BIS) from the external electronic device in accordance with the first information, using the communication circuitry, after the second information is received, detect, an event for starting an audio service provided via the communication link, in conjunction with the other electronic device, and in response to the event, transmit, to the other electronic device via the communication link, a request that ceases the synchronizing to the BIS and maintains the synchronizing of the periodic advertisement, using the communication circuitry.

[0218] For example, the at least one processor may be configured to execute the instructions to cause the electronic device to detect another event for ceasing the audio service provided via the communication link in conjunction with other electronic devices, and in response to the other events, transmit, to the other electronic device via the communication link, another request to resume the synchronizing to the BIS, using the communication circuitry.

[0219] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0220] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. s used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “**1**st” and “**2**nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” or “connected with” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0221] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of

an application-specific integrated circuit (ASIC).
[0222] Various embodiments as set forth herein may be implemented as software (e.g., the program **1340**) including one or more instructions that are stored in a storage medium (e.g., internal memory **1336** or external memory **1338**) that is readable by a machine (e.g., the electronic device **1301**). For example, a processor (e.g., the processor **1320**) of the machine (e.g., the electronic device **1301**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between a case in which data is semi-permanently stored in the storage medium and a case in which the data is temporarily stored in the storage medium.
[0223] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
[0224] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.
[0225] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

## Claims

**1**. An electronic device comprising: communication circuitry for Bluetooth low energy (BLE) communication; a speaker; memory, comprising one or more storage media, storing instructions; and at least one processor comprising processing circuitry, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via the speaker, while the first audio service is provided, receive, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a request that ceases the synchronizing to the BIS and maintains the synchronizing to the periodic advertisement, using the communication circuitry, and while providing a second audio service based on data received via the communication link from the first external electronic device, cease the synchronizing to the BIS and maintain the synchronizing to the periodic advertisement.

**2**. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: maintain the synchronizing to the periodic advertisement by receiving at least a portion of packets advertised from the second external electronic device in accordance with the periodic advertisement based on the request while providing the second audio service.

**3**. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: receive the request from the first external electronic device, through a modify source operation of a broadcast audio scan service (BASS).

**4**. The electronic device of claim 3, wherein the request indicates: maintaining the synchronizing to the periodic advertisement using a PA_sync filed of the modify source operation, and ceasing the synchronizing to the BIS, using a BIS_sync field of the modify source operation.

**5**. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: while providing the second audio service, receive, via the communication link, from the first external electronic device, another request to resume the synchronizing to the BIS, using the communication circuitry, and in response to the other request, resume to provide the first audio service by resuming the synchronizing to the BIS in accordance with the synchronizing to the periodic advertisement maintained based on the request.

**6**. The electronic device of claim 5, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: through a modify source operation of BASS, receive the other request from the first external electronic device.

**7**. The electronic device of claim 6, wherein the other request indicates resuming the synchronizing to the BIS, using a BIS_sync filed of the modify source operation.

**8**. The electronic device of claim 5, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: cease providing the second audio service, for resuming to provide the first audio service in response to the other request.

**9**. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: based on outputting audio via the speaker based on data received via the communication link from the first external electronic device, provide the second audio service.

**10**. The electronic device of claim 9, further comprising: a microphone, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: based on transmitting, via the communication link, to the first external electronic device, data regarding audio obtained via the microphone using the communication circuitry, provide the second audio service.

**11**. An electronic device comprising: communication circuitry for Bluetooth low energy (BLE) communication; a speaker; memory, comprising one or more storage media, storing instructions; and at least one processor comprising processing circuitry, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based on synchronizing, in accordance with synchronizing to a periodic advertisement of a second external electronic device indicated by information from a first external electronic device, to a broadcast isochronous stream (BIS) from the second external electronic device, provide a first audio service outputting audio via the speaker, while the first audio service is provided, receive, via a communication link between the electronic device and the first external electronic device, from the first external electronic device, a signal for starting a second audio service, using the communication circuitry, in response to the signal, cease providing the first audio service based on data for the BIS broadcasted from the second external electronic device, and based on the signal, maintain the synchronizing to the periodic advertisement, while ceasing providing the first audio service and providing the second audio service based on data received via the communication link from the first external electronic device.

**12**. The electronic device of claim 11, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: maintain the synchronizing of the periodic advertisement, by receiving at least a portion of packets advertised from the second external electronic device in accordance with the periodic advertisement, based on the signal, while providing the second audio service.

**13**. The electronic device of claim 11, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: cease providing the first audio service, by ceasing receiving the data for the BIS, in response to the signal.

**14**. The electronic device of claim 11, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: cease providing the first audio service, by ceasing rendering the data for the BIS, in accordance with the synchronizing to the periodic advertisement from the second external electronic device, using the communication circuitry, in response to the signal.

**15**. The electronic device of claim 11, wherein the signal includes call start information in accordance with a hands-free profile (HFP) or a telephone bearer service (TBS).

**16**. The electronic device of claim 11, wherein the signal includes media playback start information in accordance with an advanced audio distribution profile (A2DP) or a media control profile (MCP).

**17**. The electronic device of claim 11, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: receive, from the first external electronic device via the communication link, another signal for terminating the second audio service using the communication circuitry, while providing the second audio service, and resume providing the first audio service, by resuming the synchronizing to the BIS in accordance with the synchronizing to the periodic advertisement maintained based on the signal in response to the other signal.

**18**. The electronic device of claim 11, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: transmit, to the first external electronic device via the communication link, information indicating that the electronic device is synchronized to the BIS, using the communication circuitry.

**19**. An electronic device comprising: communication circuitry for Bluetooth low energy (BLE) communication; memory, comprising one or more storage media, storing instructions; and at least one processor comprising processing circuitry, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: transmit, to another electronic device via a communication link between the electronic device and the other electronic device, first information on a periodic advertisement of an external electronic device, using the communication circuitry, receive, via the communication link, from the other electronic device, second information indicating a state of the other electronic device synchronized to a broadcast isochronous stream (BIS) from the external electronic device in accordance with the first information, using the communication circuitry, after the second information is received, detect an event for starting an audio service provided via the communication link, in conjunction with the other electronic device, and in response to the event, transmit, to the other electronic device via the communication link, a request that ceases the synchronizing to the BIS and maintains the

synchronizing of the periodic advertisement, using the communication circuitry.
**20**. The electronic device of claim 19, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: detect another event for ceasing the audio service provided via the communication link in conjunction with other electronic devices, and in response to the other event, transmit, to the other electronic device via the communication link, another request to resume the synchronizing to the BIS, using the communication circuitry.